A-A-50964A
29 Dec 1989
SUPERSEDING
A-A-50964
13 April 1984

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINES, TOOL ATTACHMENT, PRECISION

The General Services Administration has authorized the use of this Commercial Item Description.

Abstract. This description covers precision motor driven tool post grinding machines of the standard, heavy duty, commercial type, which are used in a production environment. The machines are intended for both external and internal grinding operations when attached directly to the compound slide of a lathe or when properly attached to other machine tools.

Classification. The grinding machines shall be of the following types. The type and size to be furnished shall be as specified.

Type I - A belt driven grinding machine which mounts onto the compound slide of a lathe or similar machine by means of a T-bolt mounting post.

Type II - A direct drive grinding machine which mounts into the tool post slot of a lathe or similar machine by means of tool post shank.

SALIENT CHARACTERISTICS

1. Design. The grinding machine shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The machine shall meet the requirements specified herein and conform to current industrial practice. The machine shall be designed to perform both internal and external cylindrical grinding operations. The machine shall be convenient to operate, and adaptable to a wide variety of grinding tasks when used with lathes, planers, milling machines, boring mills, and similar machines.

2. Materials. Materials used in the manufacture of the machine shall be of a quality necessary to produce a product that conforms to the requirements specified herein.

Beneficial comments recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, TN 38114-5051.

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3. Sizes and capacities. The Type I and Type II grinding machines shall meet the size and capacity requirements of table I.

4. Construction. The grinding machine shall be so constructed that when installed and connected to the specified electrical power, it shall be ready for operation. The grinding machine structure shall withstand all forces encountered during operation of the machine, to its maximum rating and capacity without permanent distortion.

5. Measurement system. All dimensions in this description are given in US units. These dimensions may be converted to SI units through the use of conversion factors and methods specified in FED-STD-376.

6. Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

7. Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. The motor shall have a dripproof enclosure.

8. Electrical cord. Each grinding machine shall be provided with not less than six feet of flexible electrical cord, permanently attached to the machine with an appropriate electrical plug for connection to the power supply.

9. Electrical power. Grinding machines having 6 inch diameter grinding wheel and smaller shall operate on 115 volts AC, 60 Hz power. Grinding machines having 8 inch diameter grinding wheels and larger shall operate on 230 volts AC, 60 Hz, 3 phase power. An electric switch of the proper electrical rating shall be provided to turn the grinding machine on or off, and the electrical system shall be wired to conform to NFPA 79.

10. External and internal grinding. Each grinding machine shall be capable of switching from external to internal grinding operations. Each machine shall be furnished with the necessary arbors, spindle extensions, collet type chucks, or interchangeable spindles and quills to provide grinding with the size wheels listed in table I for the size grinding machine specified.

11. Drive pulleys and belts. The grinding machine shall be furnished with the manufacturer's standard set of drive pulleys and drive belts to provide all spindle speeds for which the grinder is designed.

12. Spindle. The spindle shall be accurately machined so that radial runout does not exceed .0005 inch total indicator reading (TIR).

13. Bearings. The spindle shall be mounted in ball or tapered roller bearings. All bearings (except for sealed-for-life-types) shall be provided with accessible means of lubrication.

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TABLE I. Sizes and capacities

Type I Size	Type II Size	Grinding wheel dia. max. (inches) 1/		Horsepower (Minimum)	Spindle speed RPM (no load) 2/		Standard grinding wheels	
		External	Internal		Minimum	Maximum	Straight wheels Diameter and width (inches) 3/	Mounted wheels Diameter, width and length (inches)
1	--	2	2-1/4	1/14	10,000	22,500	2-1/4	1/4x1/4x1/8, 1/2x1/4x1/8, 2x1/4x1/8
2	--	3	1-1/4	1/4	6,600	38,500	1-1/4x1/4, 2x3/8,3x3/8	1/4x1/4x1/8, 1/2x1/4x1/8, 3/4x1/4x1/8
3	--	4	1-1/4	1/3	5,500	45,000	1-1/2x1/4, 2x3/8, 3x3/8	1/4x1/4x1/8, 1/2x1/4x1/8, 1x1/4x1/4
4	--	5	1-1/2	1/2	4,200	32,600	1-1/2x1/4, 2-1/2x1/2, 4x1/2, 5x3/8	1/4x1/4x1/8, 1/2x1/4x1/8, 1x1/4x1/4
5	--	6	2-1/2	3/4	3,200	27,000	2-1, 2-1/2x3/8, 4x1/2, 5x1/2, 6x1/2	----
6	--	8	5	1	2,700	6,000	3x3/4, 5x3/4, 8x3/4	----
7	--	10	4-1/2	2	2,350	4,550	4x3/4, 4-1/2x3/8, 10x1-1/2	----
8	--	14	5	3	2,050	4,300	2x1/2, 3x3/4, 5x3/4, 12x1/2	----
9	--	14	6	5	1,700	4,300	2x1/2, 3-1/2x1/2, 5x3/4, 14x2	----
--	1	6	1-1/2	1/4	3,000	3,600	1-1/2x3/8, 6x3/8	----
--	2	8	2	1/2	3,000	3,600	2-1/2, 8x3/4	----

1/ The maximum diameter of external and internal grinding wheels shall be not greater than the figures shown in the table for the type and size grinding wheel.

2/ The spindle speed range shall be from not more than the minimum shown to not less than the maximum shown for the type and size grinding machine required.

3/ The hole diameter of the standard wheels furnished shall be the same diameter as the spindle of the type and size grinding machine required.

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14. Belt tension. Means shall be provided to adjust belt tension.

15. Belt guard. Belt driven machines shall be equipped with a belt guard.

16. Threads. All threaded parts shall conform to FED-STD-H28.

17. Wheel guard. Each grinding machine shall be provided with a metal wheel guard. The guard furnished shall meet the design, dimensions, and wheel size requirements of ANSI B7.1.

18. Wheel collars. Metal flanges or collars shall be provided with the applicable spindle, spindle extension, or quill. The flanges or collars shall conform to the requirements of ANSI B7.1 and OSHA, 29 CFR 1910. Means for securing the wheels and flanges to the work spindle shall be provided.

19. Vertical adjustment. Type I grinding machines shall be provided with a means to furnish the necessary vertical adjustment of the spindle for positioning to any lathe.

20. T-bolt mounting post. Type I grinding machines shall have the necessary T-bolt mounting post for mounting the grinding machine onto the T-slot. T-bolt dimensions shall conform to ANSI B5.1.

21. Tool post shank. Type II grinding machines shall have the necessary tool post shank to securely attach the grinding machine to the machine tool post.

22. Grinding wheels. A grinding wheel of the size specified in table I for the size grinding machine required, shall be furnished with each machine. The wheels furnished shall be general purpose, verified bond, medium grit, meeting the requirements of ANSI B7.1, ANSI B74.13, ANSI B74.2, and ANSI B74.12.

23. Diamond wheel dresser. A diamond wheel dresser shall be furnished with each grinding machine.

24. Wrench set. The manufacturer's standard set of wrenches shall be furnished with each grinding machine.

25. Tolerances. The grinding machine shall be capable of grinding the workpiece to size and roundness tolerances of 0.0005 inch, and a surface finish of 20 micro-inches.

26. Carrying case. A metal carrying case shall be furnished with each grinding machine of eight inch diameter wheel capacities, and smaller. Type I-size 4, and smaller machines may be supplied with high impact plastic case, as an option.

27. Safety devices. The machine, its components and safety devices shall conform with the requirements of OSHA, 29 CFR 1910 that are applicable to the machine itself.

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28. Painting. The grinding machine shall be painted in accordance with the manufacturer's commercial practice and color.

29. Nameplate. A nameplate shall be securely attached to each grinding machine and shall contain name of manufacturer, model and serial number, minimum and maximum wheel size specifications.

30. Workmanship. Workmanship of the grinding machine shall be of a quality equal to that of comparable products furnished to the commercial market.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Environmental protection. Under the operation, service, transportation, and storage conditions required, the item shall not emit materials hazardous to the ecological system as prohibited by federal, state, or local statutes in effect at the point of installation. The item shall be so designed as to minimize hazardous waste generated in its use or in the disposal of the item.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Responsibility for compliance. All items shall meet all requirements of salient characteristics paragraphs. The inspection set forth in this description shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

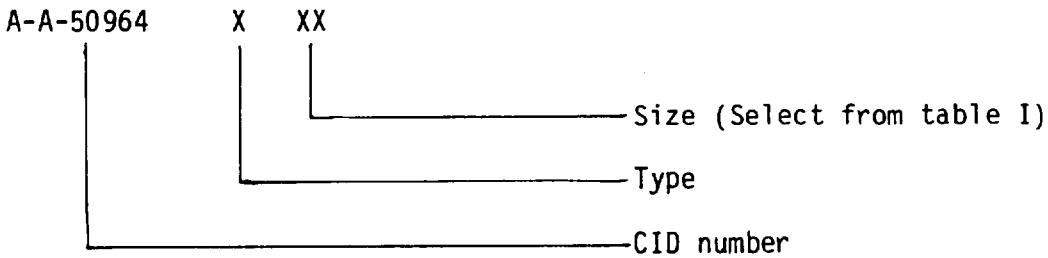
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Sample inspection. As a minimum sampling inspection, the supplier shall randomly select the number of samples in table I that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination, the test, and the inspection. Should any of the sample items fail to meet either the examination or inspection below, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all items presented at one time for acceptance.

TABLE II. Sampling plan.

Quantity of items being purchased	Sample Size	Quantity of items being purchased	Sample Size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1,200	80
26 to 50	8	1,201 to 3,200	125
51 to 90	13	3,201 to 10,000	200
91 to 150	20		

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable, Arabic numbers shall be substituted for Roman numerals.



Notes.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Type, size, and quantity required (see table I).

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c. Distance from top of lathe tool compound slide to the lathe spindle and size of mounting post T-bolt required for type I grinding machines. (These dimensions taken from lathe on which the grinding machine will be mounted).

d. Tool post slot size for type II grinding machines. (This dimension taken from machine on which grinding machine will be mounted).

Federal Specification PPP-P-40 is available from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

Federal Standard FED-STD-376 is available from the Naval Publications and Form Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

Federal Standard FED-STD-H28	- Screw Thread Standards for Federal Services.
American National Standards Institute ANSI B5.1	- T-slots, their Bolts, Nuts, Tongues, and Cutters
American National Standards Institute ANSI B7.1	- Safety Code for the Use, Care, and Protection of Abrasive Wheels.
American National Standards Institute ANSI B74.2	- Specification for Shapes and Sizes of Grinding Wheels and for Shapes, Sizes and Identification of Mounted Wheels.
American National Standards Institute ANSI B74.12	- Size of Abrasive Grain Grinding Wheels, Polishing and General Industrial Uses, Specifications for the
American National Standards Institute ANSI B74.13	- Marking for Identifying Grinding Wheels and Other Bonded Abrasives

ANSI B74.13 is available from the American National Standards Institute, Attn: Sales Department, 1430 Broadway, New York, NY 10018-3363.

American Society for Testing and Materials ASTM D 3951	- Standard Practice for Commercial Packaging.

ASTM D 3951 is available from American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.

National Fire Protection Association NFPA 79	- Electrical Standard for Industrial Machinery.

Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.

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Code of Federal Regulations (CFR) - U.S. DEPARTMENT OF LABOR, (OSHA) 29 CFR 1910 - Occupational Safety and Health Standards

Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-50964A	Grinding Machines, Tool Attachment, Precision

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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